University of Yaoundé I
Department of Physics, Faculty of Science
P.O. Box 812, Yaoundé, Cameroon
myke-vital.sao@facsciences-uy1.cm

August 25, 2026

The Editors
Journal of Computer-Aided Molecular Design
BioMed Central

Dear Editors,

We are pleased to submit our manuscript, **"Scaffold-controlled benchmarking of graph, transformer, and topological representations for antimalarial activity prediction"**, for consideration as a Research Article in the *Journal of Computer-Aided Molecular Design*.

The manuscript presents a scaffold-controlled comparison of graph, sequence, and topological molecular representations against an ECFP4–random-forest reference. We test this question on a frozen panel of 19 836 African antimalarial natural products under random and Bemis–Murcko scaffold cross-validation (25 fold–seed replicates), comparing GIN, GIN–TFP, GIN–TNE, and ChemBERTa with ECFP4–RF.

Under the scaffold split, the practically relevant scaffold-held-out setting, **ECFP4–RF achieves the highest ROC AUC (0.830 0)**, ahead of GIN–TFP (0.813 8), GIN–TNE (0.809 0), GIN (0.804 7) and ChemBERTa (0.786 7); all four learned arms are significantly lower after multiplicity correction. Under the random split the ordering is unchanged: ECFP4–RF reaches 0.943 3, versus 0.912 1 for ChemBERTa, 0.909 8 for GIN, 0.908 4 for GIN–TFP and 0.891 8 for GIN–TNE (all paired $p < 0.0001$ after Benjamini–Hochberg correction). The comparison shows that the fingerprint baseline remains strongest under scaffold-held-out testing. The topological fusion arm narrows the gap modestly, while projection-weight analysis identifies persistent-image dimensions as a salient block; this is presented as descriptive evidence about representation use, not causal attribution. The fold-independent initialization protocol is specified so that transformer performance is not inflated by cross-fold weight transfer.

As an external transfer analysis, we repeated the ECFP4–RF versus GIN comparison on a molecule-disjoint ChEMBL-derived *P. falciparum* activity panel ($n = 22\,267$ after exclusion of 180 canonical-SMILES overlaps). ECFP4–RF again outperformed GIN under the random (0.954 7 versus 0.923 7) and scaffold (0.919 0 versus 0.884 3) splits. This supports, but does not universalise, the conclusion beyond the internal screening labels; the ChEMBL labels are thresholded IC_{50}/EC_{50} records rather than prospective experiments. The study therefore offers a controlled, reproducible benchmark whose negative result is bounded to the evaluated panel, split protocols, and model configurations. Code, frozen splits and trained checkpoints are available in the public repository (`https://github.com/NanaEngo/Malaria_codesV2`).

This work is original, is not under consideration elsewhere, all authors have approved the submission, and the authors declare no competing financial interests.

We thank you for your consideration.

Myke Vital Sao Temgoua
Corresponding Author